

Comparative Nonlinear High-Sideslip Vehicle Stabilization Using PID, LQR, Gain-Scheduled LQR, Constrained MPC, and EKF State Estimation

Final Research Report - V3 Validated MATLAB Simulation Study

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Validation status: PASS. The final uploaded run passed the V3 numerical research-validation gate.

Scope: MATLAB simulation only. The results are not physical-vehicle validation and are not a real-vehicle operating guide.

This report documents the frozen final run used for the GitHub repository, including controller benchmarks, Monte Carlo robustness, EKF/MPC ablations, local control analysis, and supporting Figure-8 outputs.

Abstract

This study presents a reproducible MATLAB framework for nonlinear high-sideslip vehicle stabilization using PID, fixed LQR, gain-scheduled LQR, constrained MPC, and EKF state estimation. The validation plant is a planar four-wheel rear-wheel-drive model with individual Fiala-type tyre forces, combined longitudinal/lateral friction sharing, load transfer, resistance forces, and actuator dynamics. The research benchmark is near 32 km/h and -35 deg sideslip. Six deterministic scenarios and a matched 100-case Monte Carlo study are used. The final V3 numerical gate passed. Fixed LQR and GS-LQR pass all six deterministic scenarios and achieve 100% Monte Carlo success; constrained MPC passes three of six and achieves 87%; PID does not satisfy the predefined criterion. Under high sensor noise, EKF feedback reduces MPC sideslip RMSE from 43.587 deg to 0.221 deg (99.49% relative reduction). Conclusions are limited to the declared model, tuning, and uncertainty set.

Keywords: nonlinear vehicle dynamics, high-sideslip stabilization, LQR, gain scheduling, MPC, EKF, Fiala tyre model, Monte Carlo robustness, MATLAB

1. Research Objective and Contributions

The objective is a fair and reproducible comparison under common nonlinear dynamics, references, disturbance timing, matched uncertainty, and explicit success criteria. Controller-law performance is separated from estimator performance, and the steady research benchmark is separated from the visual Figure-8 demonstration.

- Four-wheel nonlinear validation model with per-wheel tyre saturation, friction sharing, load transfer, and actuator dynamics.
- PID, fixed LQR, GS-LQR, and constrained MPC in one reproducible framework.
- Six deterministic standardized scenarios and matched 100-case Monte Carlo uncertainty.
- Common EKF output feedback plus estimator and MPC constraint ablations.
- Local controllability, observability, pole, computation-time, and reproducibility checks.
- Separate Figure-8 F1-style visualization for communication and portfolio use.

2. Nonlinear Vehicle and Tyre Model

The validation state is $[X, Y, \psi, v_x, v_y, r, \delta_{act}, \theta_{act}]$. Front wheels steer and rear wheels receive drive force. Each wheel has its own slip angle, normal load, Fiala lateral force, longitudinal contribution, and friction utilization. Quasi-static load transfer, aerodynamic drag, rolling resistance, steering/throttle actuator dynamics, steering limits, and a physical steering-rate limit are included.

Nominal vehicle parameters

Parameter	Value	Unit
Mass	1280	kg
Yaw inertia	2100	kg m ²
CG to front axle	1.15	m
CG to rear axle	1.45	m
Track width	1.55	m
CG height	0.48	m
Nominal friction	1.12	-
Front cornering stiffness	76000	N/rad
Rear cornering stiffness	82000	N/rad
Maximum drive force	12000	N
Maximum steering	38	deg
Steering actuator time constant	0.075	s
Throttle actuator time constant	0.12	s

3. Controllers and State Estimation

PID: equilibrium feedforward plus sideslip/yaw and speed corrections.

Fixed LQR: custom CARE-based state feedback around a representative high-sideslip equilibrium.

GS-LQR: local gains interpolated across 30/35/40 km/h and absolute sideslip angles of 30/35/40 deg.

Constrained MPC: scheduled local linear model, horizon 10, 16 projected-gradient iterations, input limits, and explicit command-rate constraints.

EKF: state estimation from simulated speed, yaw rate, lateral acceleration, steering position, and throttle measurements.

4. Experimental Design

The research step is 0.02 s over 24 s; Monte Carlo uses 0.04 s. The fixed seed is 20260930. The research reference is approximately 32 km/h and -35 deg sideslip. Scenarios are dry baseline, wet-road friction drop, crosswind impulse, +15% mass/+10% inertia mismatch, high sensor noise, and a combined challenge. The same 100 randomized parametric perturbations are reused across all controllers.

5. Validated Results

5.1 Validation gate

The final numerical status is PASS. Baseline trajectories and Monte Carlo metrics are finite. Fixed LQR, GS-LQR, and constrained MPC stabilize the nominal benchmark. PID is retained as a comparison baseline.

5.2 Baseline tracking

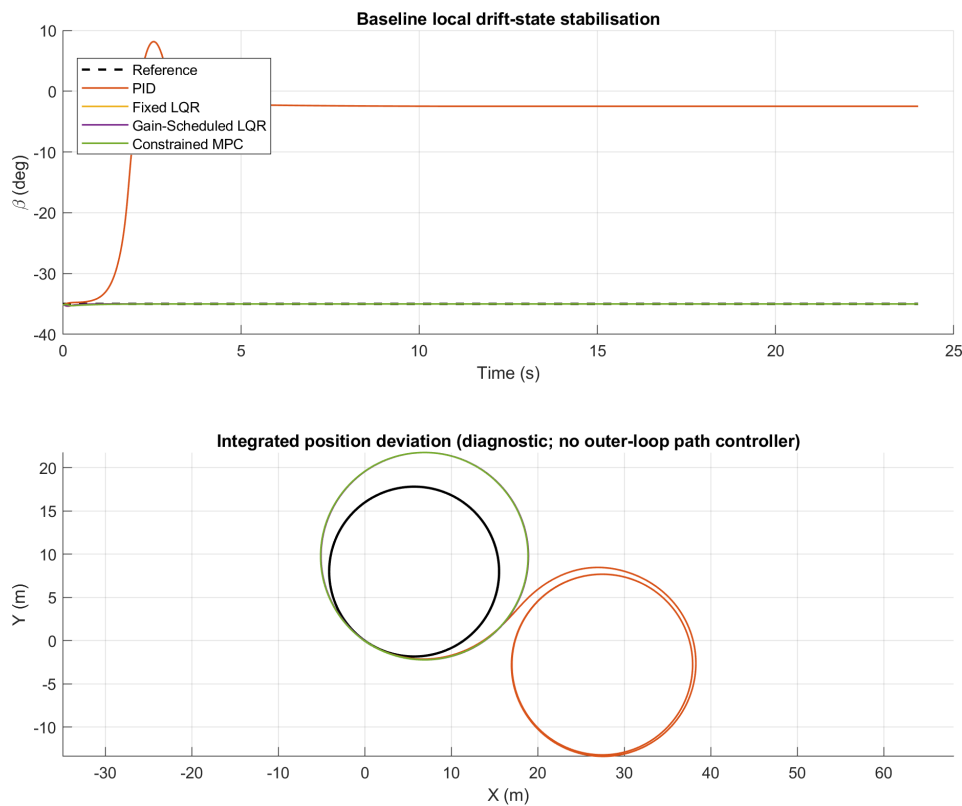


Figure 1. Baseline high-sideslip tracking for all four controllers.

Controller	Beta RMSE	Yaw RMSE	Speed RMSE	Mean ms	Result
PID	32.291118	99.678194	1.424456	0.003305	FAIL
Fixed LQR	0.001446	0.005130	0.001484	0.005494	PASS
Gain-Scheduled LQR	0.000938	0.003766	0.001021	0.024218	PASS

Controller	Beta RMSE	Yaw RMSE	Speed RMSE	Mean ms	Result
Constrained MPC	0.012860	0.013080	0.012910	0.472759	PASS

The advanced controllers stabilize the local nominal equilibrium. The extremely small LQR-family errors are consistent with a local regulation task inside the controller design envelope and are not physical-vehicle accuracy claims.

5.3 Six-scenario robustness

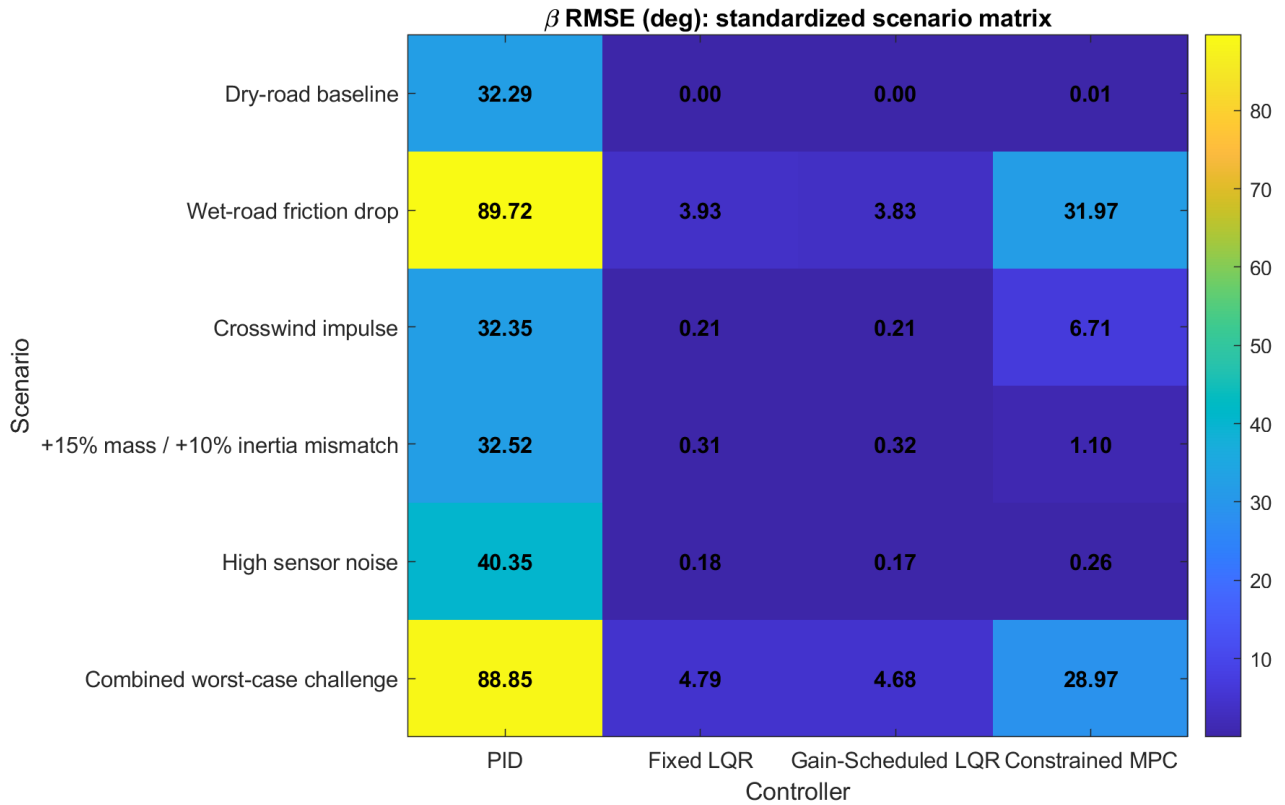


Figure 2. Sideslip RMSE across the six standardized scenarios.

Scenario	PID	Fixed LQR	GS-LQR	MPC
Dry-road baseline	32.291 deg ✗	0.001 deg ✓	0.001 deg ✓	0.013 deg ✓
Wet-road friction drop	89.716 deg ✗	3.926 deg ✓	3.834 deg ✓	31.966 deg ✗
Crosswind impulse	32.347 deg ✗	0.211 deg ✓	0.210 deg ✓	6.706 deg ✗
+15% mass / +10% inertia mismatch	32.522 deg ✗	0.314 deg ✓	0.323 deg ✓	1.097 deg ✓
High sensor noise	40.355 deg ✗	0.176 deg ✓	0.173 deg ✓	0.259 deg ✓
Combined worst-case challenge	88.850 deg ✗	4.791 deg ✓	4.676 deg ✓	28.966 deg ✗

Fixed LQR and GS-LQR pass all six scenarios. Constrained MPC passes baseline, payload/model mismatch, and high-noise cases but fails the declared criterion in wet-road, crosswind, and combined cases. PID fails all six.

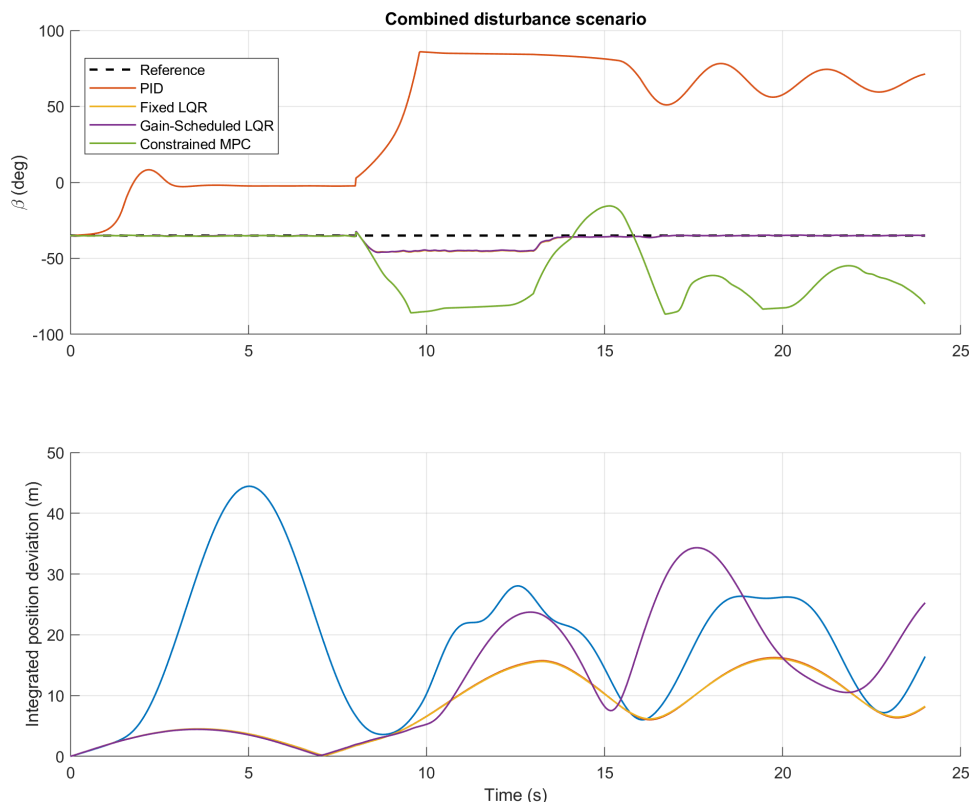


Figure 3. Combined disturbance scenario with EKF output feedback.

5.4 EKF ablation

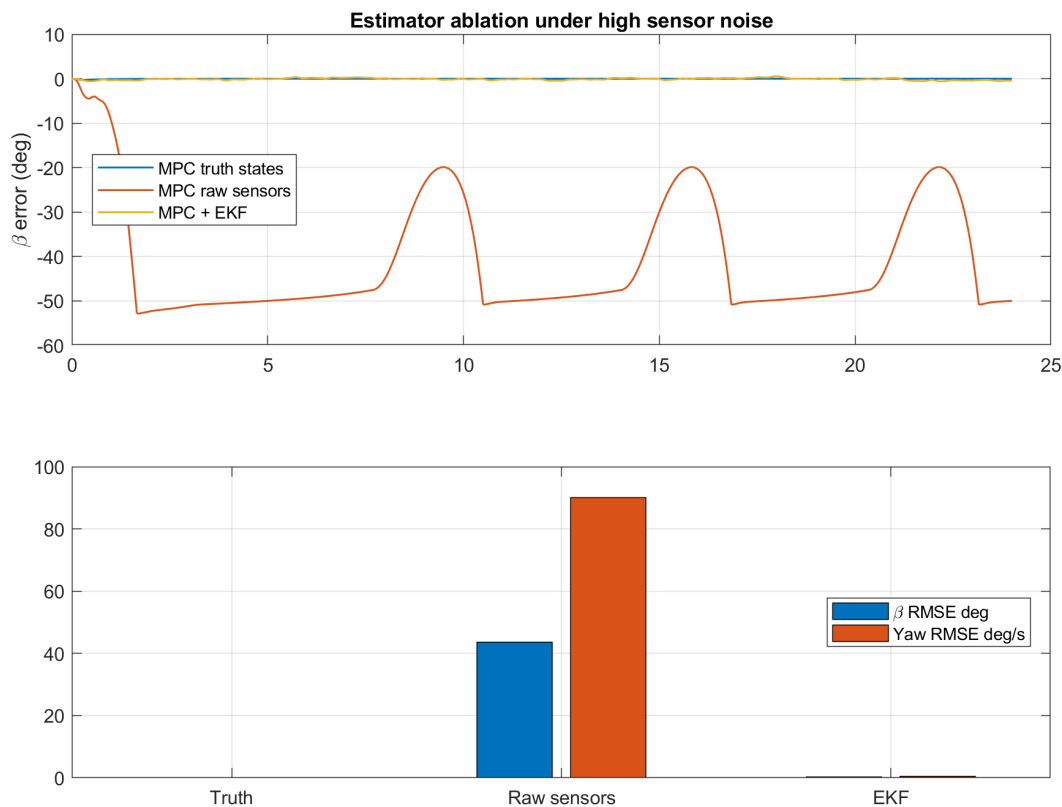


Figure 4. MPC estimator ablation under high sensor noise.

Variant	Beta RMSE	Yaw RMSE	Speed RMSE	Result
Truth states	0.012860	0.013080	0.012910	PASS
Raw sensors	43.586948	90.114104	27.789735	FAIL
MPC + EKF	0.220668	0.567298	0.185018	PASS

The EKF reduces MPC sideslip RMSE by 99.49% relative to raw-sensor feedback in the defined high-noise experiment.

5.5 MPC command-rate-constraint ablation

Variant	Beta RMSE	Position RMSE	Rate exceedances	Result
MPC no slew constraint	19.695724	16.535018	540	FAIL
Constrained MPC	30.081924	13.560719	0	FAIL

The explicit rate constraint removes raw command-rate exceedances but does not recover acceptable tracking in the severe combined challenge.

5.6 Matched Monte Carlo robustness

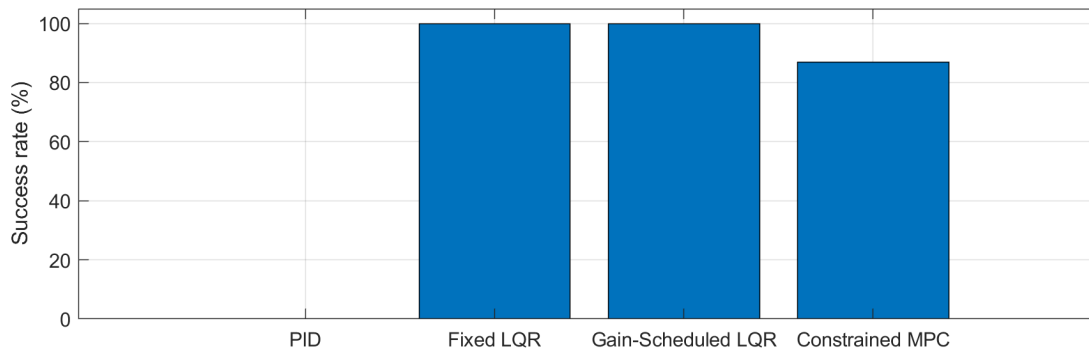
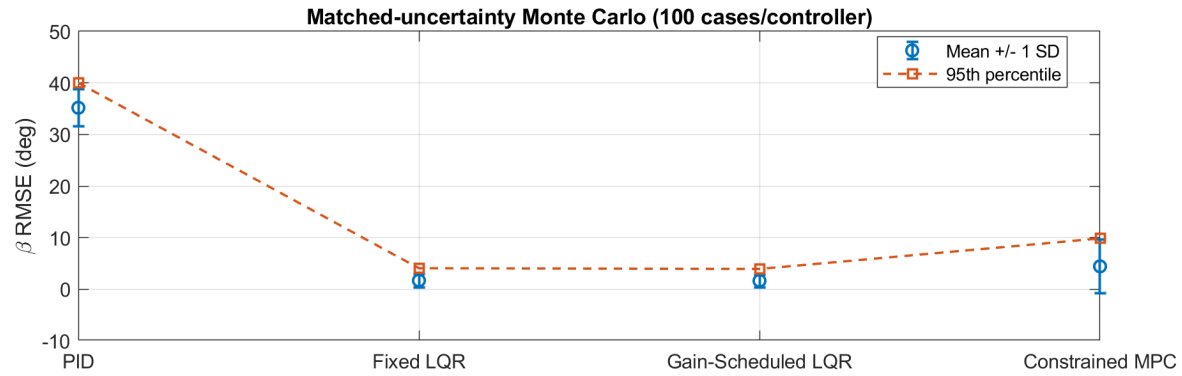


Figure 5. Matched 100-case Monte Carlo statistics.

Controller	Mean	Std.	P95	Worst	Success
PID	35.180 deg	3.590 deg	40.012 deg	40.375 deg	0%
Fixed LQR	1.690 deg	1.279 deg	4.080 deg	6.623 deg	100%
Gain-Scheduled LQR	1.640 deg	1.240 deg	3.938 deg	6.391 deg	100%
Constrained MPC	4.439 deg	5.245 deg	9.853 deg	36.695 deg	87%

Fixed LQR and GS-LQR both achieve 100% success. GS-LQR has the lowest mean, P95, and worst-case sideslip RMSE in the tested uncertainty set. Constrained MPC achieves 87% and has a heavier error tail. PID achieves 0%.

5.7 Local control analysis and computation

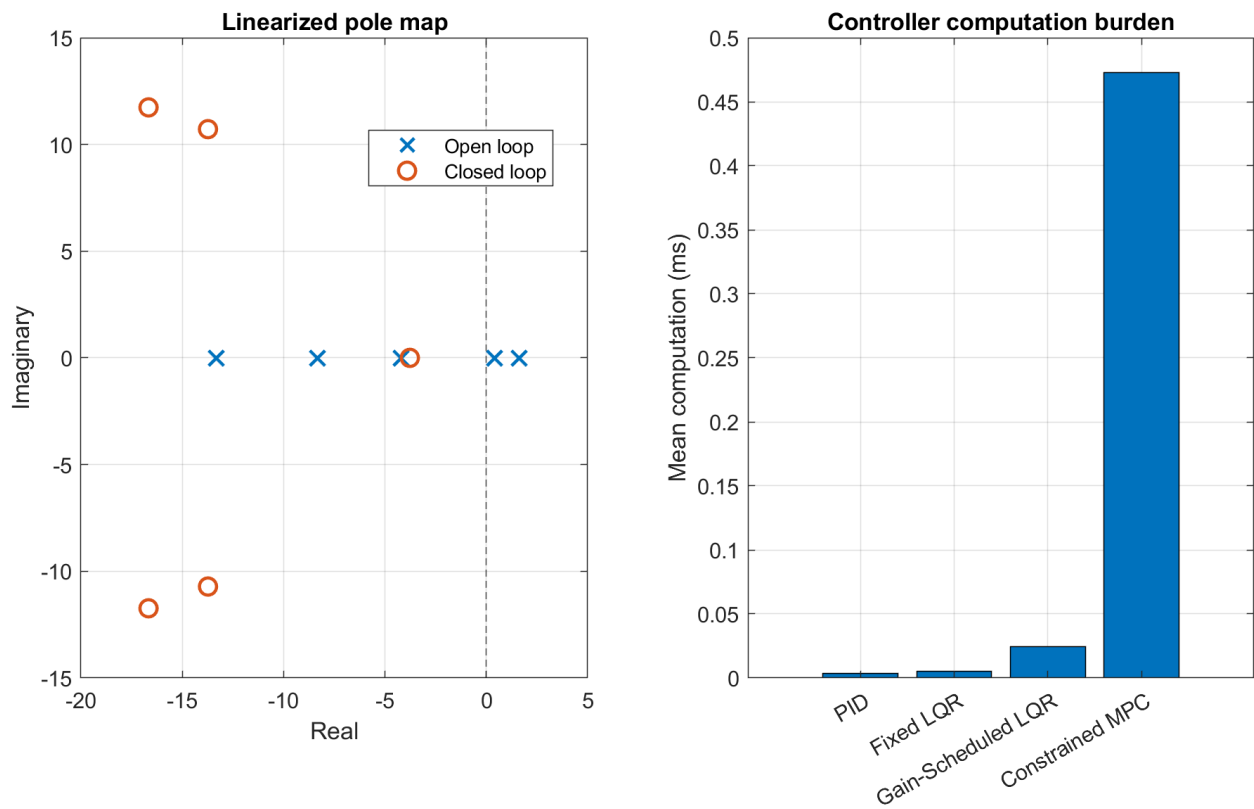


Figure 6. Local control analysis and computation-time comparison.

The five-state augmented local model has controllability rank 5/5 and observability rank 5/5. The reported fixed-LQR local closed-loop poles all have negative real parts.

Controller	Mean compute (ms)	Maximum (ms)
PID	0.003305	0.032100
Fixed LQR	0.005494	0.062100
Gain-Scheduled LQR	0.024218	0.151300
Constrained MPC	0.472759	1.119300

These timings are implementation- and hardware-dependent.

6. Supporting Figure-8 Simulation Outputs

The visual demo is separate from the research benchmark. Equilibrium-search target: 29.6 km/h and -30 deg sideslip.
 Choreographed Figure-8: 28.8 km/h nominal display speed; beta varies approximately from -40 to +35 deg.
 Saved equilibrium: approximately 9.82 m radius, -17.2 deg steering, and 35.8% throttle.

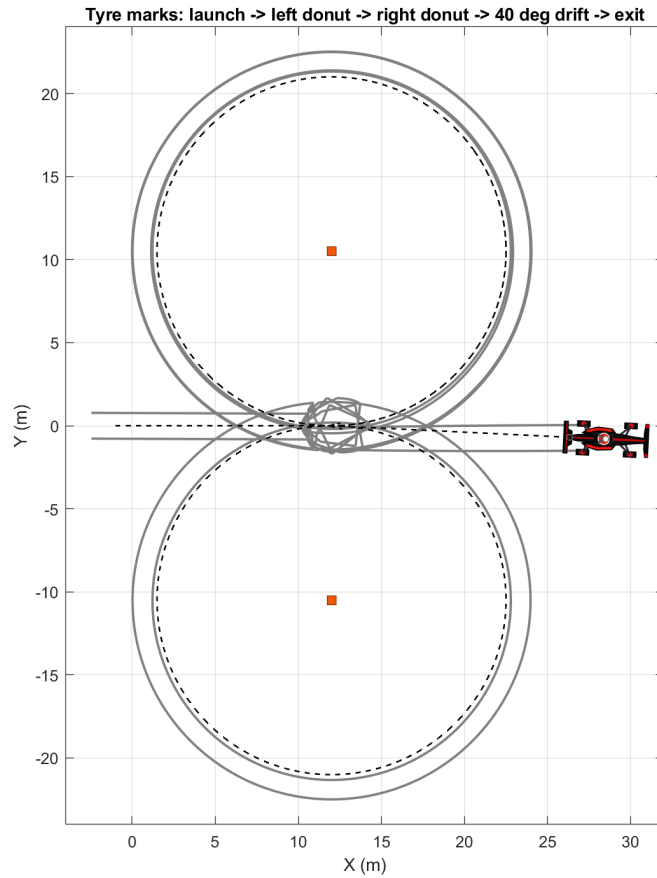


Figure 7. Figure-8 tyre-mark trajectory and final F1-style car visualization.

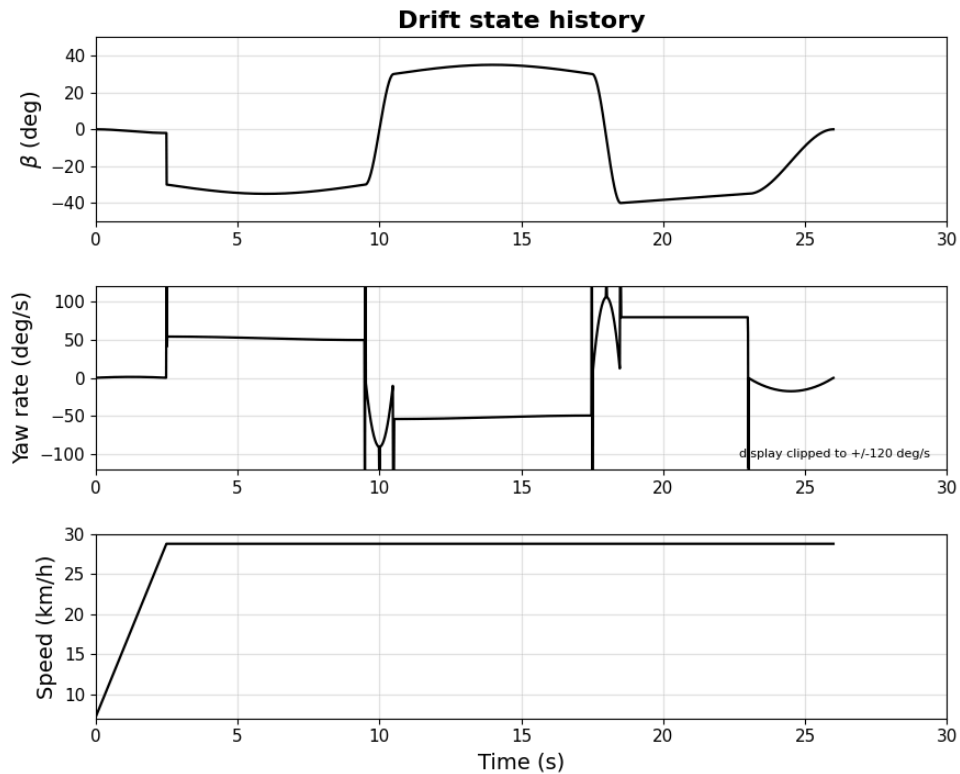


Figure 8. Drift state history: sideslip, yaw rate, and speed.

Note: the visual-demo yaw-rate panel is display-clipped to +/-120 deg/s; raw values remain in simulation_data.mat.

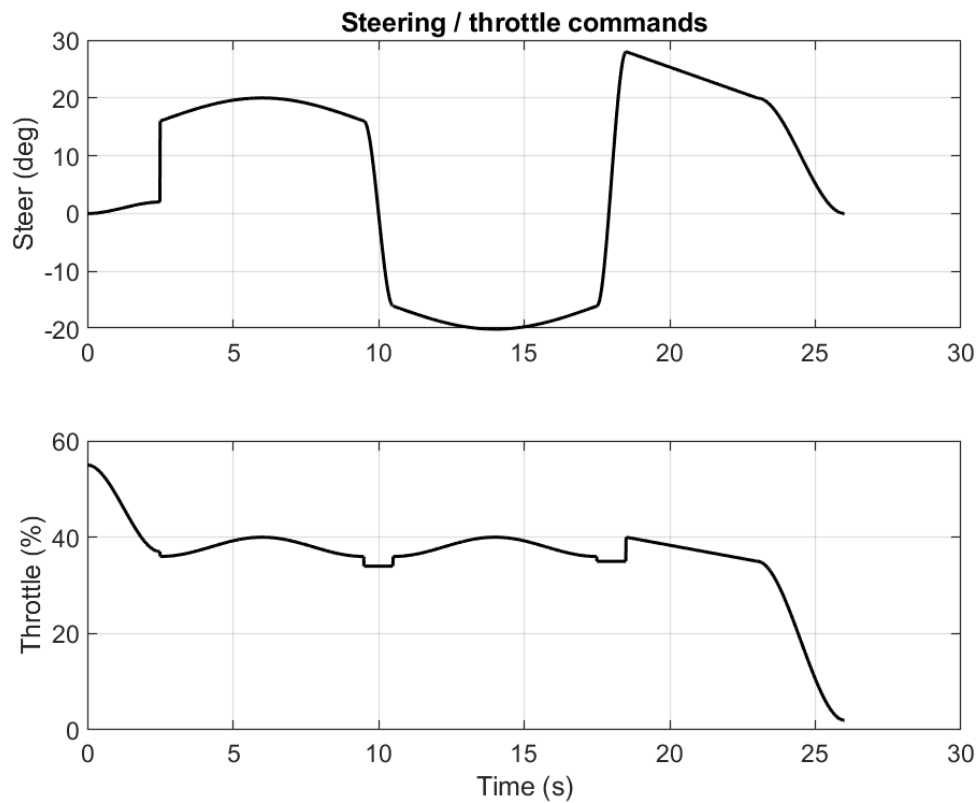


Figure 9. Steering and throttle command histories.

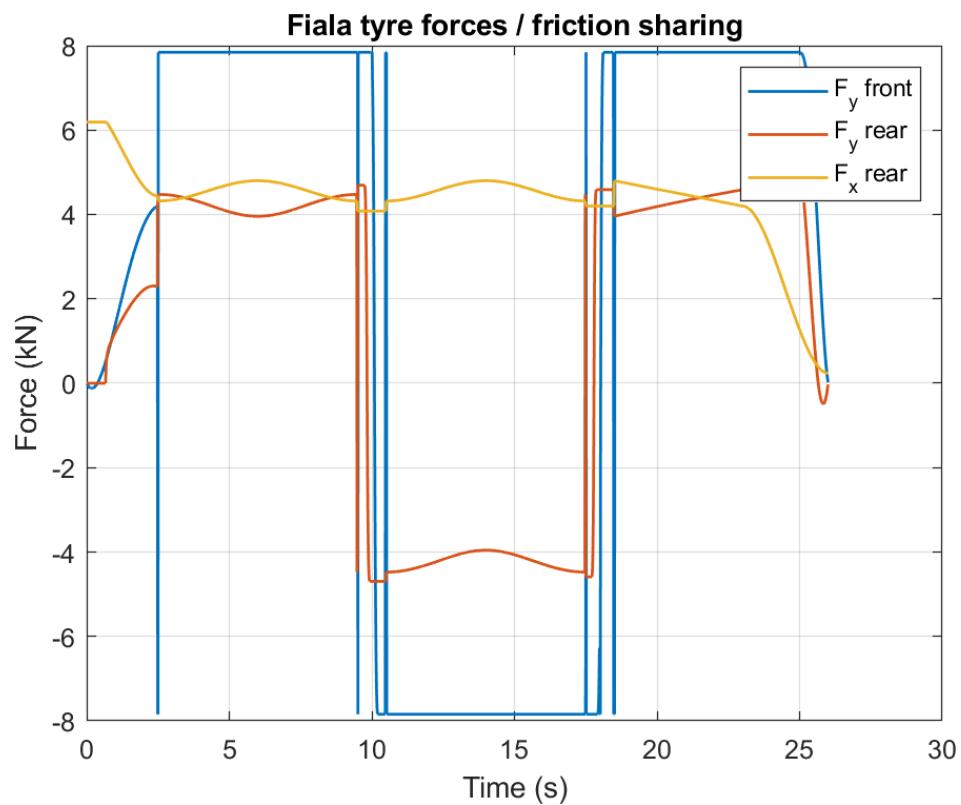


Figure 10. Fiala tyre forces and friction sharing.

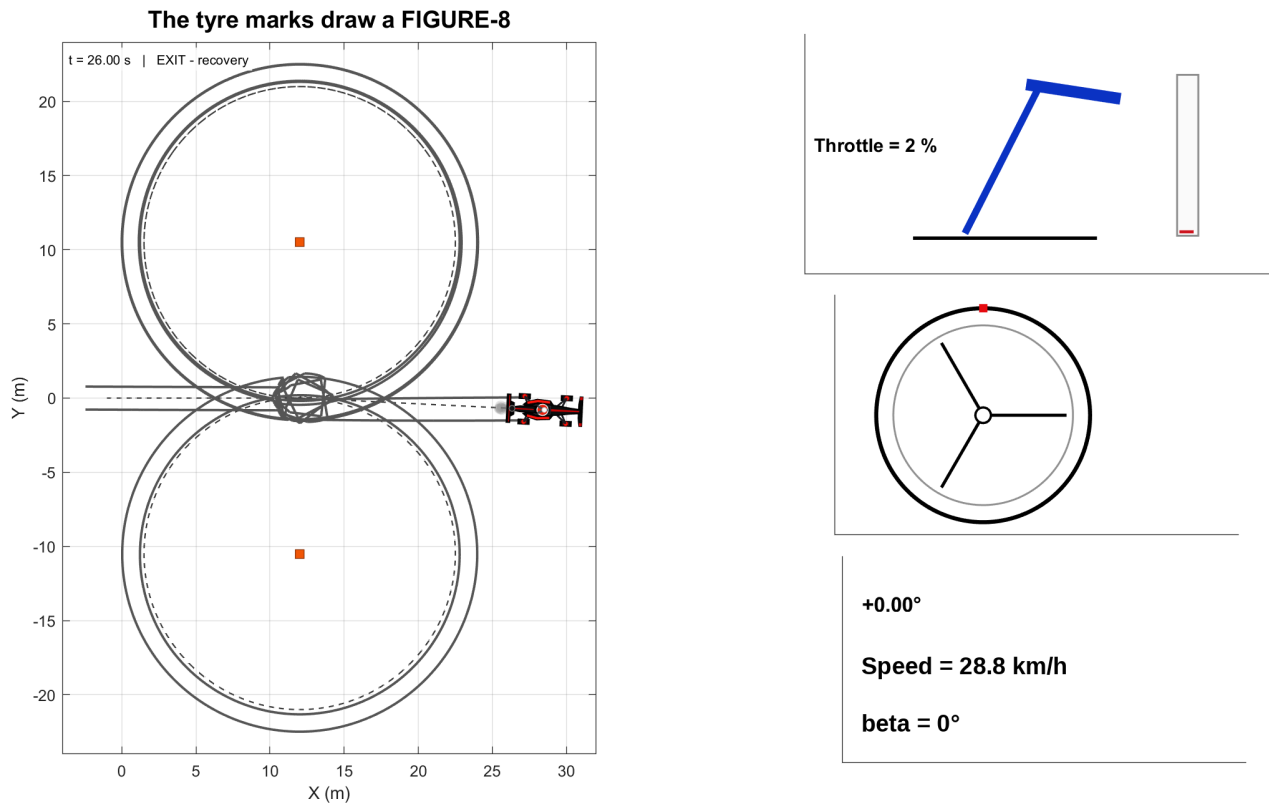


Figure 11. Final 16:9 reference visualization with HUD.

7. Discussion

The results do not establish a universal controller ranking. For this plant, tuning, and uncertainty set, fixed LQR and GS-LQR are the most consistent. The tested MPC implementation is highly accurate nominally and enforces command-rate constraints, but is more sensitive to severe friction/disturbance mismatch. The EKF ablation shows that estimator quality materially affects output-feedback performance.

8. Limitations

- Simulation only; no physical vehicle, HIL, or experimental tyre-data validation.
- Planar dynamics and quasi-static load transfer simplify real vehicle behavior.
- Local high-sideslip stabilization rather than autonomous global path tracking.
- MPC uses a local linear prediction model, finite horizon, and finite iterations.
- Controller tuning is specific to the tested envelope.
- Monte Carlo covers selected parametric uncertainty only.
- Computation times depend on MATLAB and host hardware.
- Non-MPC command-rate exceedance counts are diagnostic; physical actuator dynamics/rate limiting remain in the plant.

9. Reproducibility

Frozen final run order: `self_test -> generate_paper_results -> main -> record_github_video`. The final uploaded run produced `SELF_TEST_OK.txt`, a PASS validation gate, all research figures/CSV tables, supporting visual outputs, `simulation_data.mat`, and a playable 16:9 H.264 video. The two large V7.3 research workspace files are reproducible and intentionally omitted from the GitHub package.

10. Conclusion

The V3 project provides a reproducible nonlinear high-sideslip benchmark with four controller architectures, EKF state estimation, deterministic disturbances, matched Monte Carlo uncertainty, ablations, local control analysis, and a separate Figure-8 visualization. Fixed LQR and GS-LQR are the most robust in the tested experiment set; constrained MPC combines excellent nominal regulation and explicit rate compliance with greater severe-mismatch sensitivity; and EKF state estimation is essential for the tested high-noise MPC case.

References

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